Lesson 2.1 - How to Make a Simple Frequency Table

> **Video:** How To Make a Simple Frequency Table by The Organic Chemistry Tutor **Chapter:** 2 - Graphs and Tables

What is a frequency table?

A **frequency table** has two columns:

V (value)	f (frequency)
each different value in the data	how many times that value occurs

How to build one:

1. Find the **lowest** value and count how many times it appears.
2. Move to the **next** value up and count again.
3. Repeat until you reach the highest value. Skip values that do not appear.

Using a frequency table to find the mean, median, mode, and range

Measure	How to read it from the table
Range	highest value − lowest value
Mode	the value with the **largest frequency** (two tied → **bimodal**)
Median	the middle value; count in from both ends using the frequencies
Mean	$\bar{x} = \frac{\sum(V \cdot f)}{\sum f}$; multiply each value by its frequency, add, and divide by $n = \sum f$

Example 1

The data contains these values and counts:

V	f	V · f
3	2	6

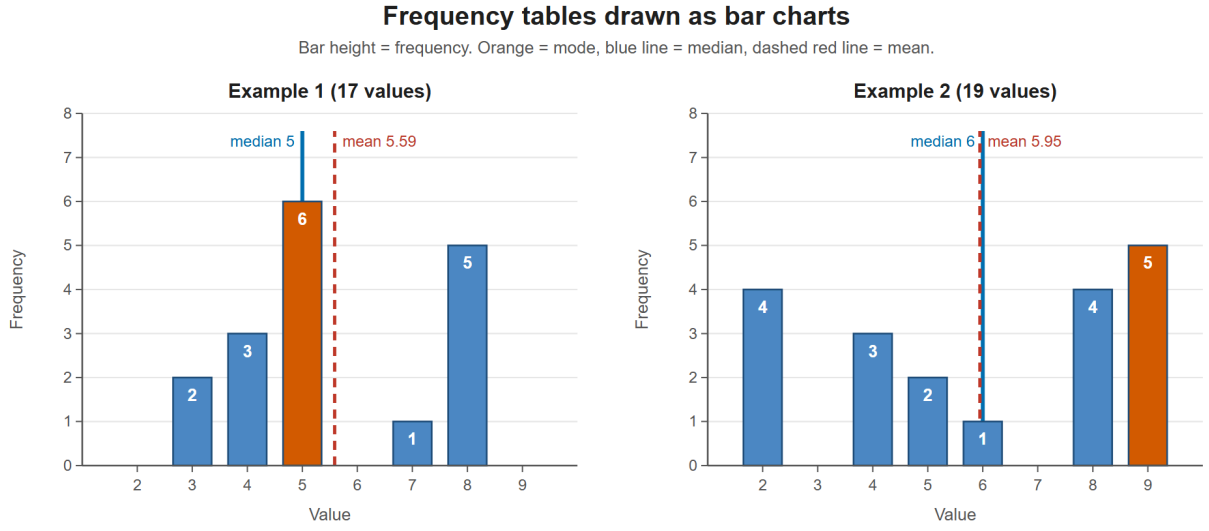
V	f	V · f
4	3	12
5	5	25
7	1	7
8	5	40

- **Range:** $8 - 3 = 5$
- **Mode:** 5 and 8 both have the highest frequency (5) → **bimodal**, modes 5 and 8.

Now add one more 5. The frequency of 5 becomes **6**:

V	f	V · f
3	2	6
4	3	12
5	**6**	**30**
7	1	7
8	5	40
Total	**17**	**95**

- **Range:** still $8 - 3 = 5$.
- **Mode:** now only **5** has the highest frequency → **unimodal**, mode = 5.
- **Median:** $n = 17$, so the median is the 9th value. Counting up: positions 1-2 are 3s, 3-5 are 4s, 6-11 are 5s. The 9th value is **5**.
  - Check by writing it out: 3 3 4 4 4 5 5 5 **5** 5 5 7 8 8 8 8 8
- **Mean:** $\bar{x} = \frac{95}{17} \approx 5.59$



Each frequency table drawn as a bar chart. The tallest bar (orange) is the mode, the solid blue line is the median and the dashed red line is the mean. In Example 1 the mode and the median are both 5. In Example 2 the mode is 9 while the median is 6, which shows that the three measures of center can disagree.

Example 2 (practice)

V	f	V · f
2	4	8
4	3	12
5	2	10
6	1	6
8	4	32
9	5	45
Total	**19**	**113**

- **Range:** $9 - 2 = 7$
- **Mode:** 9 (frequency 5, the largest)
- **Median:** $n = 19$, so the median is the 10th value. Running counts: 4 (2s), 7 (4s), 9 (5s), **10** (6). The 10th value is **6**.
  - Written out: 2 2 2 2 4 4 4 5 5 **6** 8 8 8 8 9 9 9 9 9
- **Mean:** $\bar{x} = \dfrac{113}{19} \approx 5.95$

Key takeaways

- A frequency table lists each **value** and its **frequency**.
- n = the **sum of the frequency column**.
- Total of all data = $\sum(V \cdot f)$, so **mean** = $\sum(V \cdot f)/n$.
- **Mode** = value with the highest frequency. **Range** = highest − lowest.
- **Median** = the value at position $(n+1)/2$ when n is odd, or the average of the two middle values when n is even. Use running totals of f to find it.